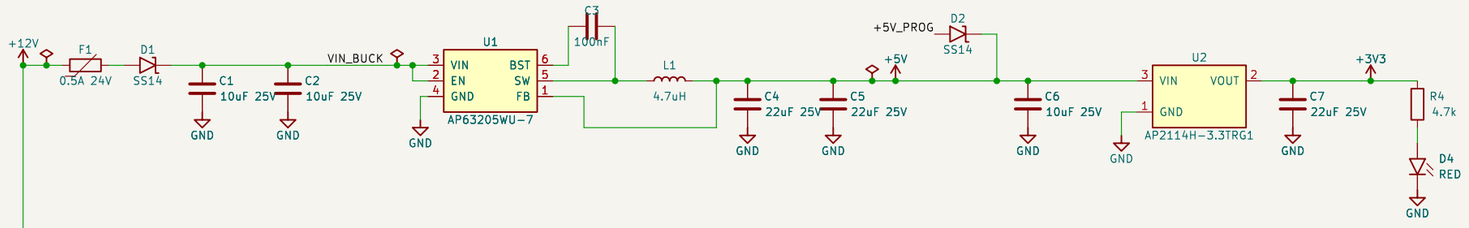
1) 12 V INPUT + BUCK 12 V -> 5 V

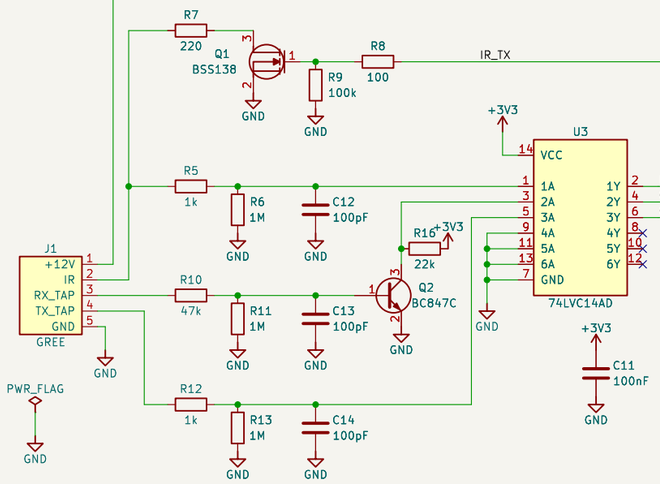
+12V and GND come from the Gree board LOGIC side through J1 (+12V pad and 'GND', never 'GND/').
F1 limits fault current into the Gree 12 V rail; D1 blocks back-feed when the board is powered from J5.



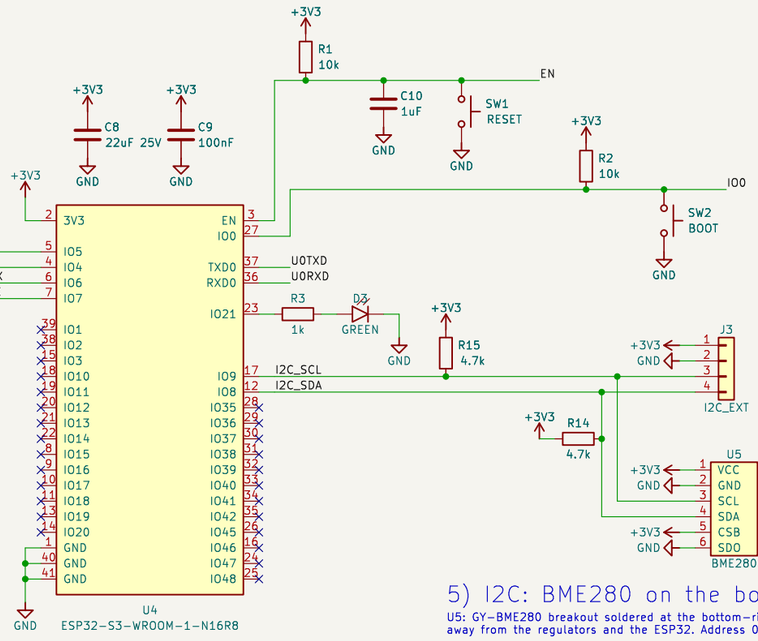
2) 5 V -> 3.3 V LDO

D2 lets J5 (USB-UART adapter 5 V) power the board on the bench.
The 5 V rail only feeds the LDO (all level shifting runs at 3.3 V).

4) IR LINE + BUS TAPS (5 V -> 3.3 V)

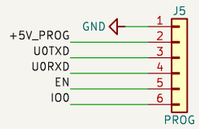


3) ESP32-S3-WROOM-1-N16R8



PROGRAMMING (3.3 V-logic USB-UART)

Labels connect by name to the same labels at the ESP32 and D2.



Hold BOOT, tap RESET to flash.
Adapter RX -> pin 3, TX -> pin 4.
After the first flash use OTA.

5) I2C: BME280 on the board + J3 for more sensors

U5: GY-BME280 breakout soldered at the bottom-right board edge, on a slotted island without copper pour, away from the regulators and the ESP32. Address 0x76 (SDO low). J3: more I2C sensors on a cable (3V3 GND SCL SDA).

J1 (one 5-pin JST XH, 2.50 mm, shared ground) on the Gree indoor board LOGIC side:
  1 +12V 2 IR data line (5 V) 3 RX tap: U6 opto output, pin 4 (0.10 V / 1.07 V); the 5 V MCU RX line also works
  4 TX tap: Gree board's TX opto (its U3) drive node, logic side 5 GND ('GND', never 'GND/')
IR line and TX tap: 1k series + 1M bleed + 100pF -> bridge U3 74LVC14 Schmitt inverter (VCC 3.3 V, 5 V tolerant inputs), ~1 Mohm load.
RX tap: 47k -> BC847 base (on above ~0.55 V, 1M pull-down, 100pF filter), 22k collector pull-up -> 74LVC14.
It takes ~10 uA from the U6 node, so the Gree board's own RX transistor keeps working normally.
Transmit: BSS138 open drain, pulls IR_LINE low only while IR_TX is high; R9 keeps it off during boot/reset.
Bridge U3 inverts: IR decoding is edge-timing based, and gree_bus.be detects the bus UART polarity (BusAuto).

ESP32: used pins sit on the side facing their circuit; every other GPIO is left open (NC flags).
Strapping pins IO0/IO3/IO45/IO46: only IO0 is used (BOOT button).
IO35/IO36/IO37 MUST stay unconnected: the octal PSRAM of the N16R8 uses them internally.

Gree GWH12KF indoor board 301484411 / GRJ819-A3 - one 5-pin connector (J1)
IR line read/inject (open drain) + indoor/outdoor bus taps + BME280

Sheet: /
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Title: Gree mini-split bridge - ESP32-S3

Size: A3	Date: 2026-10-08	**Rev: B**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1